

Calcium Vitamin D₃ STELLA

Effervescent tablet

Calcium carbonate.....	2469 mg
Calcium gluconolactate.....	100 mg
(equivalent to 1000 mg of calcium)	
Vitamin D ₃	0.022 mg
(equivalent to 880 IU of vitamin D ₃)	

COMPOSITION

Active ingredient:

One effervescent tablet contains 2469 mg of calcium carbonate and 100 mg of calcium gluconolactate (equivalent to 1000 mg of total amount of calcium) and 0.022 mg of vitamin D₃ (equivalent to 880 IU of vitamin D₃).

PHARMACODYNAMICS

- **Calcium:** Corrects an insufficient intake of calcium in the diet. The commonly accepted requirement of calcium in the elderly is 1500 mg/day.
- **Vitamin D₃:** Corrects an insufficient intake of vitamin D and increases intestinal absorption of calcium. The optimal amount of vitamin D in the elderly is 500 - 1000 IU/day.
- Administration of calcium and vitamin D₃ counteracts the increase of parathyroid hormone (PTH) which is caused by calcium deficiency and which causes increased bone resorption.

PHARMACOKINETICS

Absorption

- Calcium is absorbed from the gastrointestinal (GI) tract by active transport and passive diffusion. Calcium is actively absorbed in the duodenum and proximal jejunum and, to a lesser extent, in the more distal segments of the small intestine. The degree of absorption depends on a number of factors; calcium is never completely absorbed from the intestine.
- Vitamin D is easily absorbed in the small intestine.

Distribution and metabolism

- 99% of the calcium in the body is concentrated in the hard structure of bones and teeth. The remaining 1% is present in the intra- and extracellular fluids. About 50% of the total blood-calcium content is in the physiologically active ionised form with approximately 10% being complexed to citrate, phosphate or other anions, the remaining 40% being bound to proteins, principally albumin.
- Vitamin D and its metabolites circulate in the blood bound to a specific globulin. Cholecalciferol is converted in the liver by hydroxylation to the active form 25-hydroxycholecalciferol. It is then further converted in the kidneys to 1,25-hydroxycholecalciferol; 1,25-hydroxycholecalciferol is the metabolite responsible for increasing calcium absorption. Vitamin D which is not metabolised is stored in adipose and muscle tissues.

Elimination

- Calcium is excreted mainly in the faeces and consists of unabsorbed calcium and that secreted via bile and pancreatic juice into the lumen of the GI tract. Most of the calcium filtered by renal glomeruli is reabsorbed in the ascending limb of the loop of Henle and proximal and distal convoluted tubules. Only small amounts of the cations are excreted in urine. Calcium is also excreted by the sweat glands.
- Vitamin D is excreted in faeces and urine.

INDICATIONS

- Correction of vitamin D and calcium combined deficiency in elderly people.
- As an adjunct to specific therapy in the prevention and treatment of osteoporosis for patients who are at risk of calcium and vitamin D deficiency.

DOSE AND ADMINISTRATION

Administration

- **Calcium Vitamin D₃ STELLA** is administered orally. The tablets must be dissolved in a glass of water and the solution should then be drunk immediately after complete dissolution of the tablets.
- The tablets should be taken preferably after meals.

Dosage

- **Adult:** 1 tablet per day.
- **Dosage in hepatic impairment:**
No dose adjustment is required.
- **Dosage in renal impairment:**
This product should not be used in patients with severe renal impairment.

CONTRAINDICATIONS

- Diseases and/or conditions resulting in hypercalcaemia and/or hypercalciuria.
- Nephrolithiasis.
- Hypervitaminosis D.
- Hypersensitivity to the active substances or to any of the excipients.

PRECAUTIONS

- During long-term treatment, serum calcium levels should be followed and renal function should be monitored through measurement of serum creatinine. Monitoring is especially important in elderly patients on concomitant treatment with cardiac glycosides or diuretics and in patients with a high tendency to calculus formation. In case of hypercalcaemia or signs of impaired renal function, the dose should be reduced or the treatment discontinued.
- Vitamin D should be used with caution in patients with impairment of renal function and the effect on calcium and phosphate levels should be monitored. The risk of soft tissue calcification should be taken into account. In patients with severe renal insufficiency, vitamin D in the form of cholecalciferol is not metabolised normally and other forms of vitamin D should be used.
- Calcium Vitamin D₃ STELLA should be prescribed with caution to patients suffering from sarcoidosis because of the risk of increased metabolism of vitamin D to its active form. These patients should be monitored with regard to the calcium content in serum and urine.
- Calcium Vitamin D₃ STELLA should be used with caution in immobilised patients with osteoporosis due to the increased risk of hypercalcaemia.

- The content of vitamin D₃ (880 IU) in Calcium Vitamin D₃ STELLA should be considered when prescribing other medicinal products containing vitamin D or nutrients (such as milk). Additional doses of calcium or vitamin D increase the risk of hypercalcaemia with subsequent kidney function impairment and milk-alkali syndrome; therefore they should be taken under close medical supervision. In such cases it is necessary to monitor serum calcium levels and urinary calcium excretion frequently.

DRUG INTERACTIONS

- The effects of digitalis and other cardiac glycosides may be accentuated with the oral administration of calcium combined with vitamin D (increases the toxicity of digitalis and therefore the risk of dysrhythmia). Strict medical supervision, and if necessary, monitoring ECG and calcaemia are necessary.
- In case of concomitant treatment with a bisphosphonate or with sodium fluoride, it is advisable to allow a minimum period of two hours before taking the calcium (risk of reduction of the gastrointestinal absorption of bisphosphonate and sodium fluoride).
- Thiazide diuretics increase the renal absorption of calcium, so the risk of hypercalcaemia should be considered. Strict medical supervision of calcaemia is recommended.
- Concomitant treatment with phenytoin or barbiturates can decrease the effect of vitamin D because of metabolic inactivation.
- Concomitant use of a glucocorticosteroid can decrease the effect of vitamin D.
- Calcium salts reduce the absorption of tetracyclines. It is advisable to delay taking Calcium Vitamin D₃ STELLA by at least three hours.
- Possible interactions with food (e.g. containing oxalic acid, phosphate or phytinic acid).

PREGNANCY AND LACTATION

Pregnancy

There are no indications that vitamin D at therapeutic doses is teratogenic in humans. Calcium Vitamin D₃ STELLA can be used during pregnancy, in case of a calcium and vitamin D deficiency.

During pregnancy the daily intake should not exceed 1500 mg calcium and 600 IU vitamin D. Studies in animals have shown reproductive toxicity with high doses of vitamin D. In pregnant women, overdoses of calcium and vitamin D should be avoided as permanent hypercalcaemia has been related to adverse effects on the developing foetus.

Lactation

Calcium Vitamin D₃ STELLA can be used during breast-feeding. Calcium and vitamin D₃ pass into breast milk. This should be considered when giving additional vitamin D to the child.

ADVERSE EFFECTS

Adverse reactions are listed below, by system organ class and frequency. Frequencies are defined as: very common (≥1/10), common (≥1/100 to ≤1/10), uncommon (≥1/1,000 to ≤1/100), rare (≥1/10,000 to ≤1/1,000), very rare (≤1/10,000) or not known (cannot be estimated from available data).

Immune system disorders:

Not known: Serious allergic (hypersensitivity) reactions such as angioedema or laryngeal oedema.

Metabolism and nutrition disorders

Uncommon: Hypercalcaemia and hypercalciuria.

Gastrointestinal disorders

Rare: Constipation, dyspepsia, flatulence, nausea, abdominal pain and diarrhoea.

Skin and subcutaneous disorders

Rare: Pruritus, rash and urticaria.

OVERDOSAGE

Symptoms

Overdose leads to hypervitaminosis, hypercalciuria and hypercalcaemia. Symptoms of hypercalcaemia may include: nausea, vomiting, thirst, polydipsia, polyuria, dehydration and constipation. Chronic overdose with resulting hypercalcaemia can cause vascular and organ calcification.

The threshold for vitamin D intoxication is between 40,000 and 100,000 I.U. per day and for calcium intoxication is from supplementation in excess of 2000 mg per day, taken for several months, in persons with normal parathyroid function.

Treatment

In the case of an intoxication, treatment should be stopped immediately and the fluid deficiency should be corrected. Where overdose requires treatment, it should be via hydration, including i.v. saline solution when needed. A loop diuretic (e.g. furosemide) may then be used to further increase calcium excretion and to prevent volume overload, but thiazide diuretics should be avoided. In patients with renal failure, hydration is ineffective and they should undergo dialysis. In the case of persistent hypercalcaemia, contributing factors should be excluded, e.g. vitamin A or D hypervitaminosis, primary hyperparathyroidism, malignancies, renal failure, or immobilization.

SHELF-LIFE AND STORAGE INSTRUCTIONS

The expiry date of this pack is printed on the box. Do not use this pack after this date.

Store in a well-closed container, in a dry place. Do not store above 30°C.

Protect from moisture.

Keep out of reach of children.

PREPARATION AND PACK SIZES

Calcium Vitamin D₃ STELLA appears as a white to off-white, round-shaped, effervescent tablet, flat and plain on both sides. After reconstitution in water, tablet appears as a white solution, sweet and sour taste, orange odour with effervescence. HDPE tube of 20 tablets and packed in cardboard (packs of 20's).

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